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Capstone Project - I Synopsis (2025-26) - Sem VI

Forecasting-Driven AI System for Pollution Source Identification and Policy Decision Support

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1. Abstract

Mumbai, India's financial capital, faces severe air pollution due to vehicular emissions, industrial clusters, construction dust, shipping activities, and diesel generator usage. Despite monitoring efforts by the Maharashtra Pollution Control Board (MPCB) and SAFAR, existing systems provide only city-wide AQI updates and limited forecasts, lacking hyperlocal insights, source attribution, and policy impact analysis. This restricts effective decision-making for both citizens and policymakers.

This project proposes an AI-powered integrated and mobile platform that combines satellite imagery, MPCB monitoring data, and meteorological inputs to identify major pollution sources and forecast AQI 24–72 hours in advance. The system delivers personalized health advisories, safe route recommendations, and real-time alerts to citizens, while providing a governance dashboard for intervention impact analysis and explainable AI-based recommendations. The solution aims to shift Mumbai from reactive pollution management to predictive, data-driven action, supporting sustainable urban air quality governance.

2. Introduction

Air pollution in Mumbai is driven by a complex interaction of dense anthropogenic emissions and coastal meteorology, resulting in frequent PM2.5 concentrations exceeding WHO limits, particularly during winter inversions when reversed sea breezes trap pollutants against the Western Ghats. With over 20 million residents exposed to episodic AQI spikes often surpassing 400, the city faces serious public health risks and significant economic losses. Despite having an extensive monitoring network of 60+ MPCB stations, current systems largely provide reactive, location-agnostic data that fail to capture hyperlocal variability or identify source contributions across regions such as Bandra-Kurla Complex and Dharavi. Although recent advances in AI/ML, satellite data fusion, and meteorological modeling have achieved high forecasting accuracy, their adoption in Indian coastal megacities remains fragmented, underscoring the need for integrated, real-time platforms that combine predictive intelligence, source attribution, and decision support for both citizens and policymakers.

This gap becomes particularly evident when examining the latest peer-reviewed literature from premier publications, which reveals both impressive methodological progress and persistent systemic shortcomings :

Paper Title & Source	**Year**	**Key Methodological Contributions**	**Identified Limitations Addressed by This Work**
Comprehensive Analysis of Air Quality Trends in India (ACM DL)	2025	LSTM/RF hybrid for coastal Visakhapatnam trends	No intervention impact assessment or personalized route optimization
Forecasting India's Air Quality (IEEE Xplore)	2024	Random Forest/DT on CPCB multi-city data achieving RMSE=0.00095	Lacks source apportionment integration and policy simulation capabilities
Transforming Air Pollution Management in India (Springer Nature)	2024	Convolutional autoencoders for PM2.5 with SSIM>0.60; regulatory frameworks	Missing hyperlocal citizen interfaces and coastal meteorological adaptations
Comparative AQI Prediction: Coastal vs Non-Coastal Cities (IJERT)	2024	LSTM analysis of sea breeze effects in Mumbai/Chennai	Absence of graph-based safe routing algorithms

Drivers of PM2.5 Episodes in India (AGU/Wiley)	2024	Compositional source identification for episodic events	National scope without city-specific web/mobile deployment
Daily Scale AQI Forecasting Using Bi-RNN: Delhi Case (ScienceDirect)	2024	Bidirectional RNNs capturing temporal dependencies in station data	Station-centric approach ignores satellite and IoT fusion for comprehensive attribution
Air Quality Prediction by Machine Learning Models (ScienceDirect)	2023	CatBoost/RF ensembles yielding R^2>0.99 for nonlinear urban patterns	Black-box predictions without explainable AI for policymaker trust
Mumbai Source Apportionment & Emission Inventory (MPCB/ScienceDirect)	2023	Receptor modeling (PMF) quantifying dust (60%) and vehicles (40%)	Static annual analysis requiring real-time ML dynamism
Air Pollution Prediction: Maharashtra Case Study (PMC)	2022	ML models for SO_2/NO_x in industrial zones	Narrow sectoral focus without multi-source integration
Source Characterization of Aerosols Over Delhi (IEEE Xplore)	2020	MODIS AOD trends analysis	Pre-deep learning era lacking CNN-LSTM spatiotemporal modeling

These studies collectively validate the technical feasibility of AI-driven pollution forecasting while exposing critical deficiencies: the absence of integrated platforms combining source identification, predictive analytics, explainable policy recommendations, and user-centric applications. Most works remain siloed focusing either on methodological innovation or domain-specific analysis without bridging the translational gap to practical deployment.

The reviewed literature highlights the novelty of certain recent research efforts that adopt an integrative design, combining forecasting, source attribution, and user interfaces within unified AI-based platforms tailored to specific urban contexts. These studies move beyond standalone backend models by attempting to connect analytical pipelines with decision-support and citizen-facing systems.

Drawing from these existing approaches, the proposed project intends to design a similar integrated framework for Mumbai's coastal pollution dynamics. The system will aim to incorporate dual web and mobile interfaces to enhance accessibility of actionable air quality information. It is planned to fuse hourly MPCB pollutant data, NASA/ISRO satellite imagery, and IMD meteorological inputs into a unified analytical pipeline. Based on validated methodologies reported in prior studies, the project will attempt to utilize models such as CNN-LSTM for spatiotemporal forecasting, ensemble techniques like XGBoost with SHAP-based explainability for source estimation, and graph-based methods for hyperlocal safe-route optimization. The objective will be to translate these research-backed techniques into a practical, interpretable, and application-oriented system suitable for Mumbai's context.

Key Features of the existing platforms mentioned in the research papers:

- The platform goes beyond prediction by incorporating causal simulation.
- It can estimate the impact and return on investment of policy measures, such as odd-even vehicle schemes.
- This approach addresses the literature's call for evidence-based governance tools.
- Personalized advisory layers are provided, driven by user profiles.
- Profiles capture details such as age, health conditions, and activities like cycling or jogging.
- The platform offers context-aware recommendations, particularly useful in a city where most residents commute through pollution hotspots daily.

3. Problem Statement

Mumbai's air quality is deteriorating chronically, with AQI often crossing 300 (Very Poor) on many days each year due to combined emissions from vehicles, industries, construction dust, maritime activities, and diesel generators.Although MPCB operates 60+ monitoring stations, the data is mostly presented as city- or zone-level aggregates and fails to reflect sharp hyperlocal differences (for example, PM levels in BKC can be almost twice those in nearby Worli).
Existing forecasts are largely limited to basic 24-hour outlooks and do not adequately account for coastal meteorology such as sea breeze patterns and temperature inversions, which strongly influence pollution episodes in Mumbai.

Core problems:

- **No reliable real-time source attribution,** making it unclear whether spikes are mainly from trucks, industry, construction, or DG sets at any given time.
- **Reactive and short-horizon forecasting,** with **no robust** 48–72 hour, meteorology-aware **predictions**.
- **Lack of personalized, hyperlocal information for citizens,** including safe routes and alerts tailored to activities such as commuting, jogging, or cycling.
- **Limited capability** for BMC and MPCB **to quantitatively evaluate the effectiveness** of measures like odd–even schemes, truck bans, construction restrictions, or DG curfews.

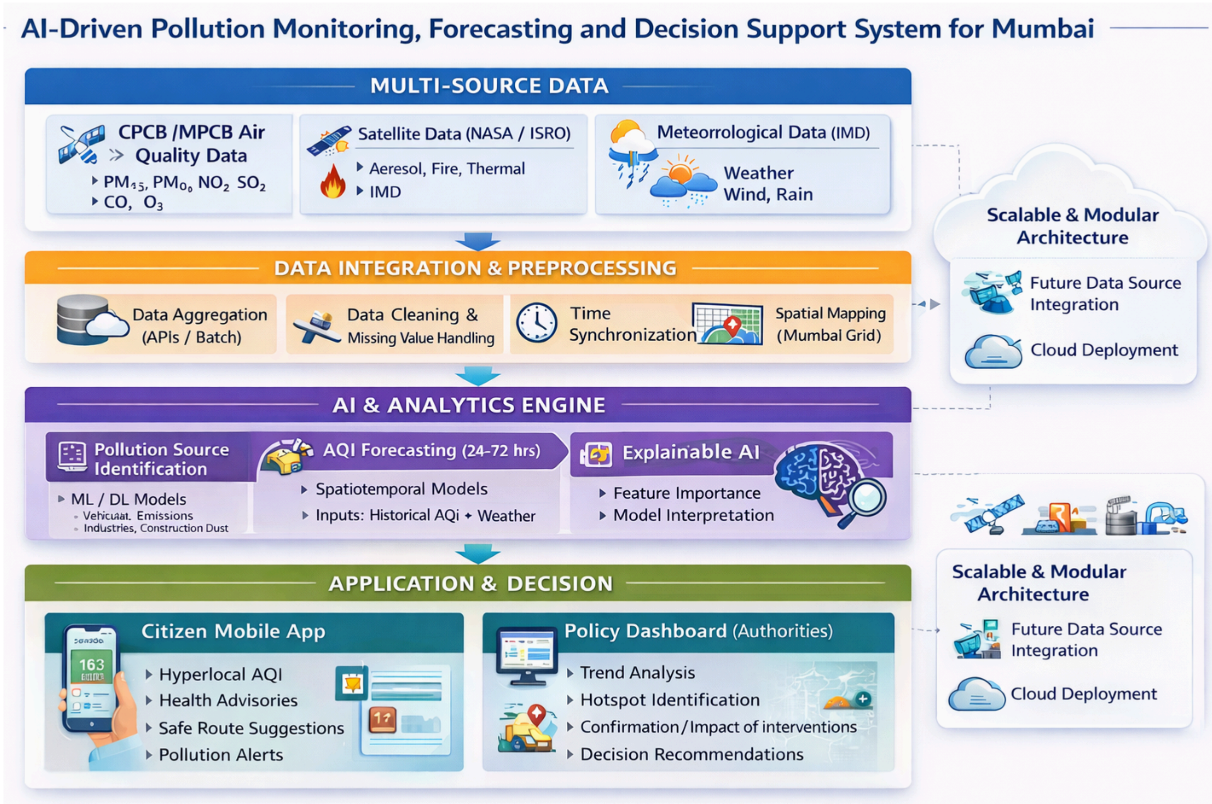
Proposed System
To develop an integrated web and mobile platform that provides real-time source contribution estimates , short- to medium-term AQI forecasts, and citizen-facing advisories with safer route suggestions.Build an administrative dashboard for BMC/MPCB that visualizes source breakdowns, tracks evolving air quality, and supports interactive exploration of different policy interventions and their projected impacts.

4. Proposed Solution

This project proposes the development of an AI-driven web and mobile platform that integrates multi-source environmental data with machine learning models to support pollution monitoring, forecasting, and decision-making in Mumbai. The system aims to shift from reactive AQI reporting to predictive, data-driven air quality management for citizens and government authorities.

Key components include:

- **Multi-Source Data Integration:**
  Aggregation of CPCB monitoring data, satellite imagery (NASA/ISRO), IMD meteorological data, and, where feasible, traffic and industrial emission data to improve pollution source estimation.
- **Pollution Source Identification:**
  ML & DL based models to estimate region-wise contributions of vehicular traffic, industries, construction dust, shipping, and DG sets.
- **AQI Forecasting:**
  Spatiotemporal models to predict AQI and pollutant levels 24–72 hours in advance using historical and meteorological inputs.
- **Citizen Application:**
  Hyperlocal AQI updates, personalized health advisories, safe route suggestions, and pollution alerts.
- **Policy Dashboard:**
  Interactive analytics for authorities to monitor trends, evaluate intervention impacts, and receive explainable AI-based recommendations.



AI-Driven Pollution Monitoring, Forecasting and Decision Support System for Mumbai
MULTI-SOURCE DATA
CPCB /MPCB Air Quality Data
PM$_{15}$, PM$_{0}$, NO$_2$ SO$_2$
CO, O$_3$
Satellite Data (NASA / ISRO)
Aeresol, Fire, Thermal
IMD
Meteorrological Data (IMD)
Weather
Wind, Rain
DATA INTEGRATION & PREPROCESSING
Data Aggregation (APIs / Batch)
Data Cleaning & Missing Value Handling
Time Synchronization
Spatial Mapping (Mumbai Grid)
AI & ANALYTICS ENGINE
Pollution Source Identification
ML / DL Models
Vehicuiat. Emissions
Industries, Construction Dust
AQI Forecasting (24-72 hrs)
Spatiotemporal Models
Inputs: Historical AQi + Weather
Explainable AI
Feature Importance
Model Interpretation
APPLICATION & DECISION
Citizen Mobile App
163
Hyperlocal AQI
Health Advisories
Safe Route Suggestions
Pollution Alerts
Policy Dashboard (Authorities)
Trend Analysis
Hotspot Identification
Confirmation / Impact of interventions
Decision Recommendations
Scalable & Modular Architecture
Future Data Source Integration
Cloud Deployment
Scalable & Modular Architecture
Future Data Source Integration
Cloud Deployment

5. Methodology

The proposed system follows a multi-stage approach consisting of data collection, preprocessing, model development, prediction, and visualization to provide real-time air quality insights and decision support.

1. Data Collection

- Collect historical and real-time air quality data (PM2.5, PM10, NO_2, CO, SO_2, O_3) from CPCB/MPCB open datasets.
- Fetch meteorological parameters (temperature, humidity, wind speed, rainfall) using OpenWeather API.
- Integrate publicly available government open datasets from the Open Government Data (OGD) Platform India for additional environmental and urban indicators.
- Store the collected data in a structured database (PostgreSQL).

2. Data Preprocessing

- Handle missing values using interpolation or forward filling.
- Remove outliers and inconsistent records.
- Perform time alignment of pollution and weather data.
- Generate additional features such as:
- Moving averages
- Lag values (previous hours/days)
- AQI calculation from pollutant concentrations.
- Normalize or scale features for model training.

3. Exploratory Data Analysis (EDA)

- Analyze pollutant trends over time.
- Identify seasonal and hourly patterns.
- Perform correlation analysis between meteorological parameters and AQI.
- Identify major factors influencing pollution levels.

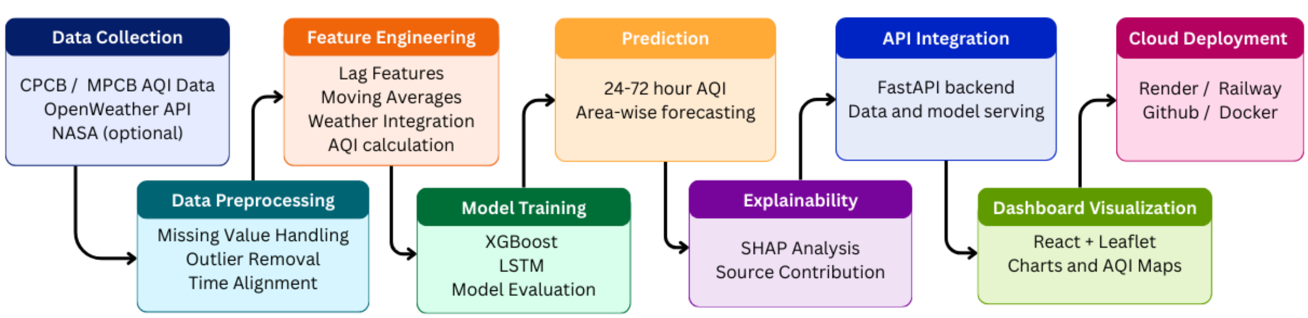


Fig.5.1: System Architecture

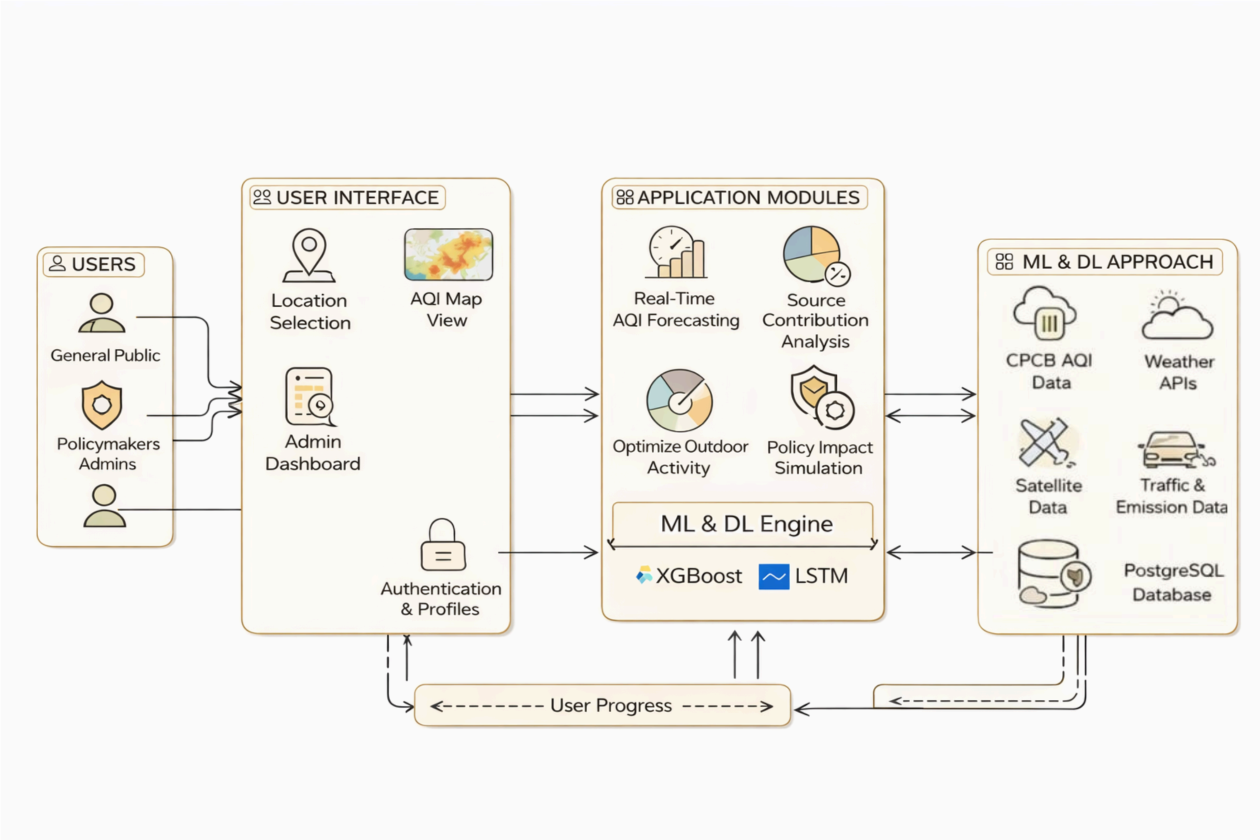


Fig.5.2: Userflow Diagram

4. Model Development

4.1 AQI Forecasting

Train XGBoost model using historical pollutant and weather data.
Use time-series features (lag values) to predict 24–72 hour AQI.
Implement LSTM model for sequential prediction.
Evaluate models using:

- RMSE
- MAE
- R^2 score

4.2 Source Contribution Analysis

- Apply SHAP (Explainable AI) on the trained model.
- Determine the relative contribution of:
  - Traffic-related pollutants
  - Industrial emissions
  - Weather influence
- Display results as percentage contribution.

5. System Integration (Backend)

- Develop REST APIs using FastAPI to:

    - Serve real-time AQI
    - Provide forecast results
    - Retrieve historical trends.
- Store processed data and predictions in the database.

6. Visualization & User Interface

Build a web dashboard using React.

Display:

- Real-time AQI map (Leaflet + OpenStreetMap)
- Forecast graphs (Chart.js)
- Pollution hotspot heatmaps
- Source contribution charts
- Provide health advisories based on AQI levels.

7. Deployment

- Train models using Google Colab.
- Deploy backend and frontend using Render/Railway.
- Maintain version control using GitHub.

6. Software and tools Requirements

Functional Features

Data & Monitoring

- **Fetch real-time AQI from CPCB** : integrate live AQI values for Mumbai stations.
- **Fetch weather data from OpenWeather** : collect temperature, humidity, wind and pressure.
- **Store historical pollution data** : maintain a time-series database for analysis and training.
- **Clean and preprocess data automatically** : handle missing, noisy and inconsistent records.

Prediction & Analysis

- **24-72 hour AQI forecasting (XGBoost/LSTM)** : generate short-term citywide AQI predictions.
- **Area-wise AQI prediction (city/zone level)** : estimate AQI for different regions of Mumbai.
- **Identify major pollution contributors using SHAP** : explain which features drive predictions.
- **Detect high-pollution hotspots** : flag zones crossing critical AQI thresholds.

Visualization (User Dashboard)

- **Live AQI map using OpenStreetMap + Leaflet** : show spatial AQI distribution on a map.
- **Color-coded AQI levels (Good–Severe)** : visually distinguish air quality categories.
- **Forecast graphs using Chart.js** : plot AQI trends for upcoming hours and days.
- **Source contribution pie chart** : visualize percentage share of different sources.

Citizen Features

- **Location-based AQI** : display current AQI for user-selected or GPS-based location.
- **Health advisory based on AQI level** : show recommended actions for each AQI band.
- **Best time for outdoor activity alerts** : suggest safer time windows for jogging or cycling.

Admin / Policy Features

- **Historical trend analysis** : inspect long-term pollution and policy impact trends.
- **Compare AQI before vs after interventions** : run basic simulations of measures taken.
- **Download reports in CSV/PDF** : export data and charts for official reporting.

Non-Functional Features

Performance

- **API response time less than 2 seconds** : ensure quick loading of dashboards and charts.
- **Hourly data updates** : refresh forecasts and visuals on a regular schedule.

Scalability

- **Cloud deployment (Render/Railway)** : host the system on scalable cloud services.
- **Modular architecture (API + ML + Frontend)** : keep components loosely coupled and extensible.

Reliability

- **Handles missing data gracefully** : avoid crashes when some sources fail.
- **Automatic error logging** : capture runtime issues for debugging and monitoring.

Usability

- **Simple UI with AQI color coding** : make the interface intuitive for all users.
- **Mobile-friendly web interface** : ensure usability on phones and tablets.

Security

- **HTTPS enabled** : protect data in transit between client and server.
- **Basic authentication for admin panel** : restrict access to administrative functions.

Maintainability

- **Clean code with GitHub version control** : track changes and collaborate easily.
- **Docker support** : containerize components for consistent deployment.

Availability

- **Cloud hosting with high uptime** : keep the service accessible most of the time.
- **Backup of historical data** : prevent loss of critical AQI and forecast history.

Software Requirements

Programming & Development

- **Python (main language)** : implement data pipeline and ML models.
- **VS Code** : use as the primary development environment.
- **Git & GitHub** : manage versions and collaborate on code.

Data & Machine Learning

- **Pandas, NumPy** : perform data cleaning and feature engineering.
- **XGBoost (main prediction model)** : build tree-based AQI forecasting models.
- **TensorFlow/Keras (LSTM)** : implement deep learning sequence models.
- **SHAP (explainable AI)** : interpret model outputs for source attribution.
- **Google Colab (model training)** : train models with GPU support online.

Backend

- **FastAPI** : expose REST APIs for data and predictions.
- **Uvicorn server** : run the FastAPI application in production.

Database

- **PostgreSQL + PostGIS** : store spatial and time-series air quality data.

Frontend

- **React** : build responsive web user interfaces.
- **Leaflet (maps)** : render interactive pollution maps.
- **Chart.js (graphs)** : display time-series and summary charts.
- **OpenStreetMap (map data)** : provide base map tiles for visualization.

Data Sources

- **CPCB AQI data** : obtain official pollution measurements.
- **OpenWeather API** : access meteorological parameters.
- **OGD India:** integrate government open datasets for environmental and urban indicators.

Deployment

- **Render or Railway** : deploy backend and frontend services.
- **GitHub** : store and manage project repositories.
- **Docker** : containerize services for easier deployment.

7. Proposed Evaluation Measures

1. Prediction Accuracy Metrics (For AQI Forecasting)

a. Mean Absolute Error (MAE)

Measures average absolute difference between actual and predicted AQI.
Lower value indicates better performance.
Formula:
MAE = (1/n) Σ |Actual − Predicted|

b. Root Mean Square Error (RMSE)

Penalizes large errors more than MAE.
Useful for measuring overall model performance.
Formula:
RMSE = √[(1/n) Σ (Actual − Predicted)^2]

c. R^2 Score (Coefficient of Determination)

Measures how well the model explains the variance.
Value range: 0 to 1
Closer to 1 = better model fit.

2. Classification Performance

If AQI categories are predicted (Good, Moderate, Poor):

- Accuracy
- Precision
- Recall
- F1 Score

Visualization & Usability Evaluation

- Correct AQI category mapping
- Real-time dashboard loading time
- User-friendly interface testing

8. Conclusion

The reviewed studies demonstrate the effectiveness of machine learning and deep learning models such as XGBoost, CatBoost, RNN, and LSTM in accurately predicting air quality using pollutant, meteorological, satellite, and emission data. However, most existing approaches remain limited to offline analysis and lack real-time deployment, dynamic source attribution, geospatial visualization, and decision-support capabilities for citizens and policymakers. To address these gaps, future work should focus on developing an integrated, scalable platform that enables real-time, hyperlocal air quality forecasting with explainable AI and policy impact assessment. Additionally, the system can be extended to incorporate **noise pollution and water pollution monitoring**, enabling multi-domain environmental analysis and supporting holistic urban sustainability and public health management in complex metropolitan regions such as Mumbai.

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